KASHAF FATIMA

AI Engineer | Agents & Evals | Founder
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SUMMARY

I build AI agents that act on real systems, and the evals that catch them being confidently wrong before anyone else does. I raised a production assistant's accuracy from **43% to 82%**, shipped a voice agent with **0 silent failures in 57 tests**, and founded Occuris AI, whose agents caught **87 of 100** planted supply problems in a live pilot.

EXPERIENCE

Occuris AI, Founder & Lead AI Engineer | Karachi (hybrid) — Jan 2026 – Present

- Caught **87 of 100** planted supply problems (95% CI 79–92%) across **300+ items** in a six-month pilot with NKE Technica, by building procurement and inventory agents (LangGraph, FastAPI) and tuning them on a known-answer set to prefer false alarms over misses.
- Made agents safe to run unattended on schedules and events, with **every irreversible action** held for human sign-off, by routing those actions through Occuris Echo for approval over Gmail and Slack.
- Kept the platform free of vendor lock-in, with every answer traceable to its model, by running agents interchangeably on Claude, Grok and Ollama and logging the model on each response.
- Made tenant isolation a database guarantee, not an app promise, by enforcing row-level security on every table from the verified Supabase Auth session.
- Lead a team of three: set the technical direction, run technical hiring, and pitched Occuris at ITCN Asia, Karachi (Sep 2026).

ProForecast, AI/ML Engineer | UK company (remote) — Sep 2024 – Mar 2026

- Raised PROPILOT's answer accuracy from **43% to 82%**, measured on an eval set I built from real Sage 50 client queries and scored on intent and the figure returned, by fixing semantic-search and intent failures with BERT/RoBERTa embeddings.
- Delivered **12-month forecasts** to clients in Germany, the US and beyond, by building RAPIER's pipeline over 18 months of client finance and warehouse data on HDFS, Spark and Snowflake.
- Took PROPILOT into French with LoRA tuning and shipped every change to both products (Django, DRF, TensorFlow) through PR review.

10Pearls, Data Science Intern (Shine) | alongside Occuris — Jul 2026 – Sep 2026

- Cut Karachi air-quality forecast error (RMSE) from **8.63 to 7.74** (R^2 0.21 to 0.36), by adding known-future weather to a three-day forecaster that hourly GitHub Actions pipelines retrain and promote across seven model families, with no manual step.

PROJECTS

Occuris Echo, governed voice agent | Python, Claude, Gmail, Calendar, Slack, Sheets — Sep 2026

- Passed **57/57** known-answer cases with **0 silent failures** (3 with the deterministic layer off), by settling requests with rules before Claude, asking instead of guessing, confirming before anything irreversible, and blocking duplicate sends.
  github.com/kashaff-codes/occuris-echo | Loom demo

Supabase Lite Conformance Kit, differential testing | TypeScript, PostgreSQL, SQLite, CI — Oct 2026

- Traced **48 failures to 22 root causes** and fixed two upstream (passing 33 to 36, silent wrong data 15 to 12, nothing else moved), by diffing 81 supabase-js calls across Postgres and SQLite and citing the PostgREST (Haskell) and GoTrue (Go) source behind each.
  kashaff-codes.github.io/supalite-conformance | upstream PR

om-toolload, agent memory benchmark | TypeScript, Mastra, Ollama, Claude — Oct 2026

- Raised Mastra Observational Memory from **17 to 29 of 38** correct ($p = 0.008$) at under a third of full context's tokens, by pinning tool values through its transform hooks, measured with a pre-registered benchmark and an audited LLM judge.
  github.com/kashaff-codes/om-toolload | demo video

supabase-ai-eval, eval-first RAG agent | Python, FastMCP, pgvector, Claude — 2026

- Made answer quality measurable on every run, through LLM-as-judge scores logged to Postgres and a live dashboard, by building hybrid HNSW and full-text retrieval served to other agents as MCP tools.
  github.com/kashaff-codes/supabase-ai-eval | live dashboard

PUBLICATION

A Known-Answer Audit for Deployed Measurement Systems: Procedure and a Case Study. Sole author, Zenodo preprint, 2026. Planted ground truth in my own deployed pipeline, found two components silently returning fallback constants for about 70% of observations, and withdrew the headline claim when bootstrap confidence intervals did not support it.
doi.org/10.5281/zenodo.22395202

SKILLS

Evals & statistics: known-answer and differential testing, LLM-as-judge, regression gates in CI, bootstrap and binomial confidence intervals, McNemar tests

Agents & LLMs: LangGraph, MCP, RAG, pgvector, human-in-the-loop approval, Claude, Grok, OpenAI, Gemini, Ollama, BERT, PyTorch, TensorFlow

Engineering: Python, TypeScript, FastAPI, Django, React, PostgreSQL, Supabase RLS, Rust, Spark, Snowflake, AWS, pytest, Vitest, Playwright, GitHub Actions; Go and Haskell (source reading)

EDUCATION

Bahria University Karachi, BS Computer Science (AI/ML specialization) — Jan 2027